Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/27/2021
PI Name:	Levine, Benjamin D M.D.		
Project Title:	Integrated Cardiovascular (ICV) 2.0: Assessing the Risk for Atrial Fibrillation in Astronauts During Long Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	benjaminlevine@texashealth.org	**Fax:**	FY 214 345-4618
PI Organization Type:	UNIVERSITY	**Phone:**	214-345-4619
Organization Name:	The University of Texas Southwestern Medical Center at Dallas		
PI Address 1:	Institute for Exercise and Environmental Medicine (IEEM)		
PI Address 2:	7232 Greenville Ave, Suite 435		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75231-5129	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health-OMNIBUS
Start Date:	07/02/2015	**End Date:**	06/30/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2017 (originally 7/1/2016) per NSSC information (Ed., 8/9/16)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Bungo, Michael M.D. (University of Texas Medical School at Houston) Opondo, Mildred M.D. (University of Texas Southwestern Medical Center at Dallas) Sarma, Satyam M.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX15AP25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Integrated Cardiovascular (ICV) Experiment was perhaps the most ambitious and comprehensive cardiovascular study performed in the International Space Station (ISS) era. Although many questions regarding the cardiovascular adaptation to long duration spaceflight were answered, a number of other key questions were raised that we propose to address in this omnibus application. The focus of this application is with the risk of atrial fibrillation (afib), the most common clinical arrhythmia in the astronaut demographic. With the aging of the astronaut population, numerous cases of afib have been observed, primarily in active astronauts on the ground. Therefore afib has become an issue of increasing importance for flight medicine. In addition, it may be that the extensive exercise performed on the ISS, which prevents cardiac atrophy and cardiovascular (CV) deconditioning, might actually increase the risk of atrial fibrillation as it does in competitive athletes. Because of the extensive cardiac structural and arrhythmia analysis already performed by the ICV investigators, only minor modifications to the ICV analysis plan would allow high resolution assessment of afib risk. We propose to analyze our already acquired data to assess atrial morphology, electrophysiology, and risk for afib in completed ICV subjects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study showed that even 6 months of atrial distension from spaceflight (compared to the upright position on Earth) can lead to both mechanical (atrial dilation) and electrical (reduction in RMS (root mean square) voltage of the terminal portion of the atrial electrogram)) changes in the atria, though the left atrial (LA) appears more susceptible to the right atrial (RA). This study adds to the accumulating knowledge that physiologic cardiac remodeling can occur in the atria from a number of different stimuli such as exercise training, or failure to reduce atrial pressure from routine upright posture during the day. These stimuli are phenotypically similar and give clues as to why athletes may have a greater risk of atrial fibrillation than non-athletes.
Task Progress:	This study has been completed and published. All 13 astronauts from the ICV study had their cardiac MRIs re-analyzed for atrial volume and function, and their Holter monitors re-analyzed for signal averaging of the P wave of the ECG. This analysis, which was the primary focus of the task, showed that the left atrial volume increased by 12 ml (~20%) after 6 months in space; in contrast, the right atrial volume did not change. P wave signal averaging revealed a progressive loss of the root mean square voltage of the terminal 20 ms of the P wave suggesting an electrical as well as mechanical remodeling. All changes were restored to baseline values upon recovery. Although no atrial fibrillation (afib) was observed in any astronaut, and most astronauts had no significant changes in the detection of arrhythmia burden, one of 13 had a very large increase in supraventricular beats, from an average of ~500/48 hours (with large day-to-day variability of the pre-flight measures) to >5,000/48 hours.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
Articles in Peer-reviewed Journals	Negishi K, Borowski AG, Popovic ZB, Greenberg NL, Martin DS, Bungo MW, Levine BD, Thomas JD. "Effect of gravitational gradients on cardiac filling and performance." J Am Soc Echocardiogr. 2017 Dec;30(12):1180-8. https://doi.org/10.1016/j.echo.2017.08.005 ; PubMed PMID: 29056408 , Dec-2017
Articles in Peer-reviewed Journals	Khine HW, Steding-Ehrenborg K, Hastings JL, Kowal J, Daniels JD, Page RL, Goldberger JJ, Ng J, Adams-Huet B, Bungo MW, Levine BD. "Effects of prolonged spaceflight on atrial size, atrial electrophysiology, and risk of atrial fibrillation." Circ Arrhythm Electrophysiol. 2018 May;11(5):e005959. https://doi.org/10.1161/CIRCEP.117.005959 ; PMID: 29752376 , May-2018